Automated Defect Detection in Aircraft Engine Turbine Blades Using Deep Learning

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Abstract

Aircraft engines operate under extreme thermal and mechanical stresses, where even minor defects such as cracks, dents, or burns on turbine blades can escalate into severe damage if undetected. Current borescope inspections rely heavily on human experts, making the process time-consuming, subjective, and prone to errors. This project proposes the development of a convolutional neural network (CNN) system to classify and localize turbine blade defects.

1 Introduction

Aircraft engines operate under extreme thermal and mechanical stresses, where even minor defects such as cracks, dents, or burns on turbine blades can escalate into severe damage if undetected. Current borescope inspections rely heavily on human experts, making the process time-consuming, subjective, and prone to errors under poor imaging conditions. Developing an automated defect detection system has the potential to significantly improve aviation safety while reducing costly downtime during maintenance checks. This project aims to train a convolutional neural network (CNN) to classify and localize common turbine blade defects, including cracks, dents, burns, and defect-free cases. Leveraging transfer learning with pretrained models such as ResNet and YOLOv5 further enhances performance while reducing training time, aligning directly with advanced deep learning techniques.

2 Illustration / Figure

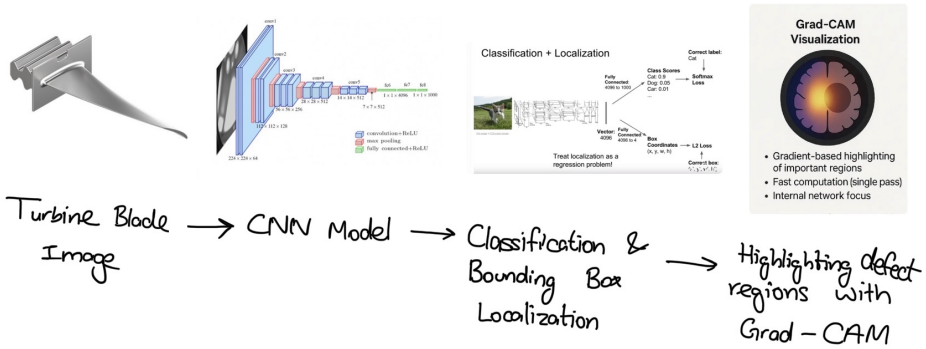

Figure 1: Pipeline sketch: turbine blade image → CNN model → classification and bounding-box localization → Grad-CAM highlighting. *Image attributions (left to right):* [1, 2, 3, 4].

3 Background & Related Work

The **AeBAD dataset (2023)** provided one of the first large-scale aero-engine blade anomaly collections, containing around 5,500 real and synthetic images and serving as a benchmark for CNN-based defect detection [5]. The **BladeSynth dataset (2025)** addressed the challenge of limited real-world data by generating realistic synthetic defects, giving researchers a valuable tool for training and validation under diverse conditions [6]. Beyond datasets, researchers have adapted deep learning architectures for borescope inspections. Zhang *et al.* [7] modified YOLOv5 with deformable convolutions, improving accuracy on complex turbine defect images (code available on GitHub). U-Net variants have been applied for segmentation tasks, with improved performance in noisy or UAV-based imaging conditions [8]. Faster R-CNN and related detection networks have been applied to aero-engine blade inspections, reporting high classification accuracy for cracks, burns, and dents [9].

This project builds on these efforts by combining both real and synthetic datasets, testing how well models generalize across domains, and incorporating interpretability tools like Grad-CAM heatmaps to provide visual explanations of predictions—an important step toward trust and usability in aviation safety applications.

4 Data Processing

This project combines several datasets to ensure sufficient scale and diversity. The AeBAD dataset provides around 5,500 labeled images of cracks, dents, burns, and normal blades [5], while BladeSynth contributes high-quality synthetic defect images [6]. Smaller Roboflow datasets (50–100 labeled images each) are also included [10]. Together, these resources exceed the 1,000+ image requirement and cover both real and synthetic domains. Images are normalized and resized to model-specific dimensions (224×224 for ResNet, 416×416 for YOLO). The data is split into training (70%), validation (15%), and testing (15%). Augmentation techniques such as random rotations, brightness/contrast adjustments, and Gaussian blur are applied to simulate challenging borescope conditions. Label categories are standardized (e.g., merging "nick" with "dent") for consistency. Training will be carried out in PyTorch using cross-entropy loss for classification and YOLO loss for localization, with optimizers such as Adam or SGD. Models will be trained for 20–30 epochs with early stopping to avoid overfitting, and batch sizes will be selected to balance efficiency and stability.

5 Architecture

A simple CNN and a Multilayer Perceptron (MLP) serve as baselines, trained directly on raw pixel values or HOG features. The MLP will include one or two hidden layers with ReLU activations and a softmax output layer. The main models build on transfer learning. A ResNet50 pretrained on ImageNet will be fine-tuned for defect classification, while YOLOv5 pretrained on COCO will be adapted for localization [7]. Model performance will be evaluated using accuracy, precision, recall, and F1-score for classification, and mean Average Precision (mAP) with Intersection-over-Union (IoU) for localization. Grad-CAM and Score-CAM will provide interpretability, highlighting image regions most influential in predictions.

6 Ethical Considerations

A key ethical concern is the risk of false negatives—missed defects could have severe safety consequences if the system were relied upon operationally. For this reason, the model should be positioned strictly as *decision support*, complementing but never replacing certified inspectors.

Another limitation stems from the reliance on synthetic data. While synthetic images expand training diversity, they may not perfectly capture the appearance of real-world defects [6]. To mitigate this, both real and synthetic data will be incorporated, and performance will be reported separately on real-world test subsets. Explainability methods like Grad-CAM will also help ensure inspectors can validate that the model is focusing on the correct regions.

References

[1] "Aerospace: Turbine blade," https://tungaloy.com/industries/aerospace_turbine-blade/, 2025.

[2] "Different types of cnn architectures explained with examples," https://vitalflux.com/different-types-of-cnn-architectures-explained-examples/, 2025.

[3] "Object detection: A gentle introduction," https://www.guidetomlandai.com/object-detection/, 2025.

[4] S. Laouadi, "Comparing grad-cam and occlusion sensitivity for medical image interpretation," https://medium.com/@dr.saad.laouadi/comparing-grad-cam-and-occlusion-sensitivity-for-medical-image-interpretation-a-brain-tumor-a2036878f88d, 2023.

[5] T. Zhang and L. Gao, "Aebad: A benchmark dataset for aero-engine blade anomaly detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*.

[6] "Bladesynth dataset: Synthetic dataset for turbine blade defect detection," Available at: https://springernature.figshare.com/articles/dataset/BladeSynth_A_High-Quality_Rendering-Based_Synthetic_Dataset_for_Aero_Engine_Blade_Defect_Inspection/28658603?file=53208893, 2025, accessed: September 27, 2025.

[7] Y. Zhang, H. Liu, and J. Wang, "A deformable convolutional yolov5 for aero-engine blade defect detection," *IEEE Access*, 2022, code available at: https://github.com/zhangzilongc/MMR, accessed: September 27, 2025.

[8] B. Liu and C. Xu, "An improved u-net model for segmenting wind turbines from uav-taken images," *Energies*, 2022.

[9] M. Zhao and Q. Chen, "Deep learning-based defects detection of certain aero-engine blades and vanes with ddsc-yolov5s," *Journal of Nondestructive Evaluation*, 2022, accessed: September 27, 2025.

[10] "Turbine blades automatic defect computer vision model)," Roboflow, https://universe.roboflow.com/shanghai-jiaotong-university-bfxbs/turbine-blades-automatic-defect, 2024.